

TECHNOLOGICAL UNIVERSITY (THANLYIN)
DEPARTMENT OF INFORMATION TECHNOLOGY ENGINEERING

**A PERSONALIZED RECOMMENDATION SYSTEM FOR
HOTEL BOOKING USING CONTENT-BASED FILTERING**

BY
MA THAZIN NYUNT
VI IT - 4

GRADUATION THESIS

OCTOBER, 2025
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**A THESIS
SUBMITTED TO THE DEPARTMENT OF
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IN PARTIAL FULFILMENT OF THE REQUIREMENTS FOR
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(INFORMATION TECHNOLOGY ENGINEERING)**

OCTOBER, 2025

THANLYIN

TECHNOLOGICAL UNIVERSITY (THANLYIN)
DEPARTMENT OF INFORMATION TECHNOLOGY ENGINEERING

We certify that we have examined, and recommend to the University Steering Committee for Bachelor of Engineering Graduation thesis entitled: **“A PERSONALIZED RECOMMENDATION SYSTEM FOR HOTEL BOOKING USING CONTENT-BASED FILTERING”** submitted by **MA THAZIN NYUNT**, Roll No. **VI IT - 4 (NOVEMBER, 2024)** to the Department of Information Technology Engineering as the requirement for the Bachelor of Engineering degree, **BE(IT)**.

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ABSTRACT

This study develops a personalized hotel booking system incorporating a content-based filtering recommendation engine to enhance user experience in online reservations. The system was developed using the MERN stack (MongoDB, Express.js, React, and Node.js) to ensure scalability, responsiveness, and usability for both customers and administrators. Core functionalities include secure user registration and authentication via Clerk, hotel and room listings with advanced filtering options, booking management, payment processing through Stripe, and automated email notifications using Nodemailer.

The main contribution of this work is the integration of a personalized recommendation system. For non-registered users, the platform displays top-rated hotels based on popularity and customer ratings to provide relevant suggestions without prior interactions. For registered users, the content-based filtering algorithm dynamically constructs user profiles by analyzing historical booking data, preferred amenities, price ranges, and ratings. Using similarity measures, the system ranks hotels and recommends those most closely aligned with each user's preferences. Additionally, an administrative dashboard supports hotel and booking management with role-based access for administrators and hotel owners.

Overall, the proposed system demonstrates that combining personalized recommendation techniques with secure and efficient booking functionalities can substantially enhance user satisfaction, decision-making, and operational efficiency in hotel reservation platforms.

TABLE OF CONTENTS

	Page
ACKNOWLEDGEMENTS	i
ABSTRACT	ii
TABLE OF CONTENTS	iii
LIST OF FIGURES	vi
LIST OF TABLES	vii
 CHAPTER TITLE	
1 INTRODUCTION	1
1.1. Introduction	1
1.2. Problem Statement	1
1.3. Aim and Objectives	2
1.4. Scope of the Thesis	2
1.5. Outline of the Thesis	2
2 BACKGROUND THEORY	3
2.1. Hotel Booking Systems	3
2.1.1. Components and Core Functions	3
2.1.2. Importance in Travel and Tourism Industry	4
2.2. Recommendation Systems	4
2.2.1. Types of Recommendation Systems	5
2.2.2. Key Algorithms and Techniques	7
2.2.3. Advantages and Limitations of Each Type	7
2.3. Content-Based Filtering	8
2.3.1. How Content-Based Filtering Works	8
2.3.2. Advantages of Content-Based Filtering	9
2.4. Personalized Recommendations in Hotel Booking	9
2.5. Related Theories and Technologies	10
2.5.1. MERN Stack Technologies	11
2.5.2. Authentication and Security with Clerk	12
2.5.3. Payment Integration with Stripe	12
2.5.4. Email Services with Nodemailer	13
2.6. Technologies Used in Implementation	13

	2.7. Summary	14
3	DESIGN OF THE SYSTEM	15
	3.1. Overall System	15
	3.1.1. Personalized Hotel Room Booking Process for Registered Users	15
	3.1.2. Hotel and Booking System Management	17
	3.1.3. Use Case Diagram for User	18
	3.1.4. Use Case Diagram for Admin	19
	3.1.5. Class Diagram	20
	3.2. Recommendation Algorithm Flow	21
	3.2.1. User Profile Construction	22
	3.2.2. Candidate Hotel Selection	22
	3.2.3. Similarity Measures	23
	3.2.4. Weighted Scoring	24
	3.2.5. Ranking and Recommendation	24
	3.3. Summary	25
4	IMPLEMENTATION OF THE SYSTEM	26
	4.1. System Overview	26
	4.2. User and Admin Registration and Authentication	27
	4.2.1. Registration Page	27
	4.2.2. Login Page	28
	4.3. Role of User	29
	4.3.1. Home Page	29
	4.3.2. Hotel Listings Page	30
	4.3.3. Hotel Details Page	31
	4.3.4. Booking Summary Page	33
	4.3.5. Stripe Checkout Page	34
	4.4. Role of Admin	35
	4.4.1. Admin Dashboard	35
	4.4.2. Add Hotel Page	36
	4.4.3. Hotel Listings Page	37
	4.4.4. Room Listings Page	37
	4.4.5. Add Room Page	38
	4.4.6. Room Availabilities Page	39

	4.5. Recommendation System Implementation	40
	4.5.1. Recommendation System Testing	41
	4.6. Summary	42
5	DISCUSSION AND CONCLUSION	43
	5.1. Discussion	43
	5.2. Further Extension	43
	5.3. Conclusion	44
	REFEREN	
	CES	45